Cold-Chain Fulfillment Center — Capacity Scaling (unexpected demand increase)

ENG1M-0000001 · Industrial Engineering · Strategy Consulting Hub

Decision

Option C — Comprehensive architecture replacement for cold-chain fulfillment center — delivers 13% capacity scaling improvement (target 8.5%) for CAD $676,000 of the CAD $866,000 budget: NPV CAD $2,681,053 at 9.5%, payback 11.1 months.

Why this option

For the cold-chain fulfillment center operated by a metropolitan public agency, plan a capacity upgrade while protecting service quality following unexpected demand increase. Baseline capacity: 1,190 units/day. Deliver at least 8.5% improvement within 3 months under CAD 866,000 capital limit; installation restricted to weekends. Select a technically justified alternative and test what happens if performance falls 15%.

Opt	Alternative	Capex	Gain	In budget	Meets target	NPV	Payback	Score /10
A	Instrumentation and operating procedure changes for cold-chain fulfillment center	CAD $232,000	4.6%	yes	no	CAD $957,263	10.8 months	4.6
B	Selective retrofit of critical cold-chain fulfillment center subsystems	CAD $502,000	6.9%	yes	no	CAD $1,252,328	14.8 months	5.26
C ✔	Comprehensive architecture replacement for cold-chain fulfillment center	CAD $676,000	13%	yes	yes	CAD $2,681,053	11.1 months	8.55

Economics

Capex	NPV @ WACC	IRR	Payback	Benefit/Cost	Run-rate benefit	Levelized cost
CAD $676,000	CAD $2,681,053	119.2%	11.1 months	4.33	CAD $947,537/yr	5.43 CAD per added unit

Bear case (performance -15%, capex +20%): NPV CAD $1,996,871, payback 14.9 months.

Plan

- **Horizon 1 (Month 1)** — Baseline validation & pilot, CAD $135,200. KPI: Pilot reaches >= 2.5% improvement; data quality verified
- **Horizon 2 (Month 2)** — Scale-up: Comprehensive architecture replacement for cold-chain fulfillment center, CAD $405,600. KPI: >= 6.8% improvement sustained for 4 weeks
- **Horizon 3 (Month 3)** — Stabilise, optimise & hand over, CAD $135,200. KPI: Full target 8.5% reached; O&M within CAD $33,800/yr

Top judge questions

Q: What changes when unexpected demand increase persists and the selected design underperforms by 15%?
Bear case (performance -15%, capex +20%): 11.0% improvement, NPV CAD $1,996,871, payback 14.9 months. Target is still met.

Q: How does your recommendation change if performance falls 15% and costs rise 20%?
Bear case (performance -15%, capex +20%): 11.0% improvement, NPV CAD $1,996,871, payback 14.9 months. Target is still met.